PARUL UNIVERSITY

CERTIFICATE

This is to certify that Project - 1 (303105300) of 6th Semester entitled **"Neural Authentication Verification and Synthetic Media Forensics System"** of Group No. **CSE_67** has been successfully completed by:

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Panchal Jaykumar S

under my guidance in partial fulfillment of the Bachelor of Technology (B.Tech) in Computer Science & Engineering of Parul University in Academic Year **2025–2026**.

Date of Submission :.....

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Neural Authentication Verification and Synthetic Media Forensics System

A Project Report

Submitted By

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in Partial Fulfilment For the Award of

the Degree of

BACHELOR OF TECHNOLOGY

COMPUTER SCIENCE & ENGINEERING

Under the Guidance of

Prof. Ankita Verma

Assistant Professor

PARUL UNIVERSITY

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April - 2026

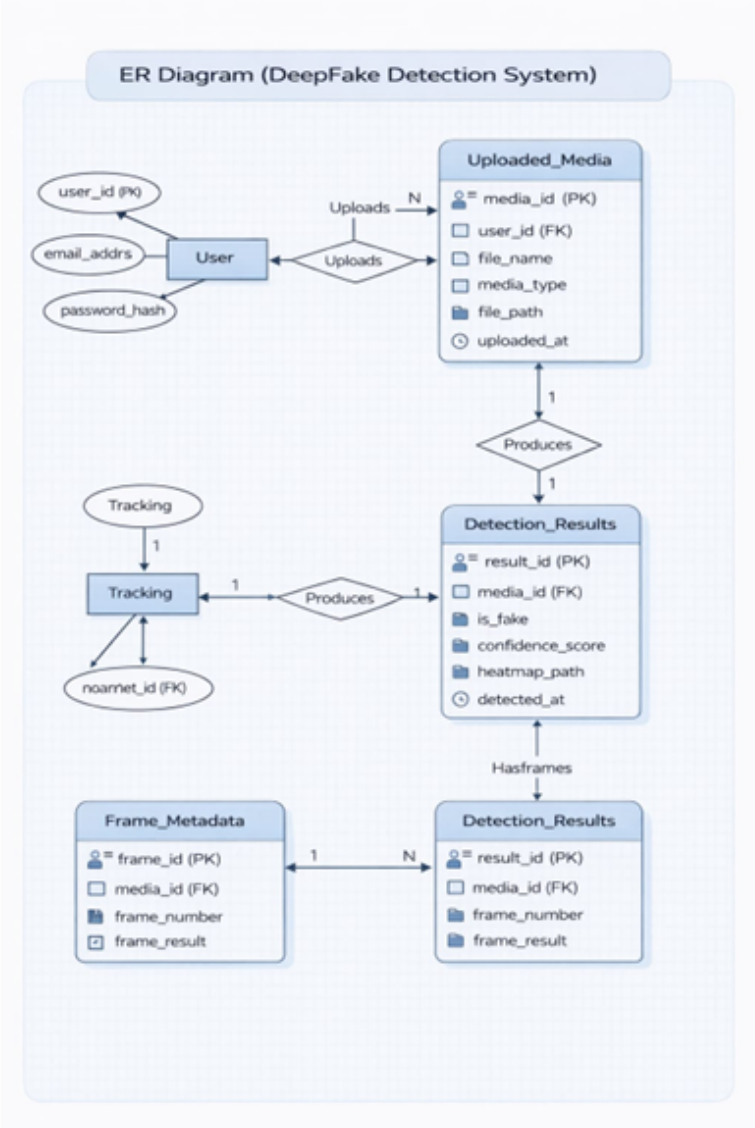

Figure 4.1: ER Diagram of DeepFake Detection System

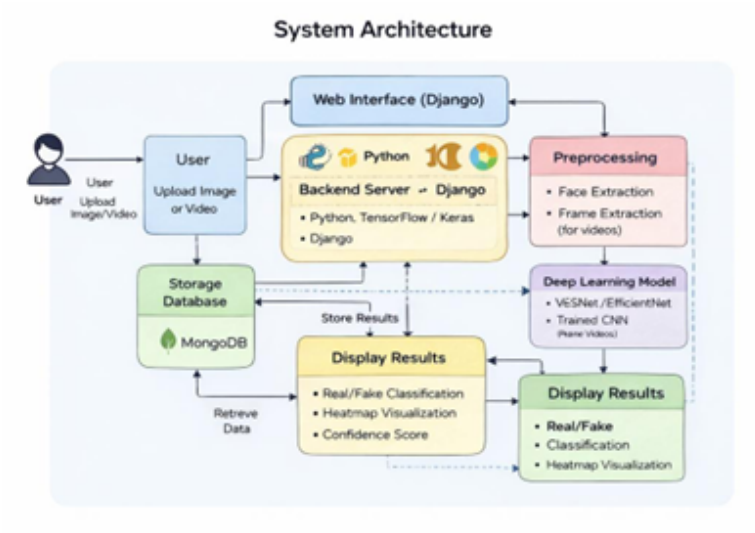

Figure 4.2: System Architecture of DeepFake Detection System

4.2 User Interface Design

The user interface is designed to be simple, intuitive, and user-friendly. The goal is to ensure that even non-technical users can easily interact with the system.

The main interface includes an upload section, where users can select and upload image or video files. Supported formats include JPG, PNG, and MP4. Once the file is uploaded, the system processes the input and displays the results.

The output section shows:

- The classification result (Real or Fake)
- Confidence percentage
- Heatmap visualization highlighting manipulated regions

The interface is designed to provide quick feedback with minimal waiting time. Clear labels and instructions are provided to guide users throughout the process. The design focuses on accessibility, ensuring compatibility across different devices and browsers.

Figure 4.3: User Interface Design

4.3 Database Design

Table	Column	Data Type
User	user_id	INT PRIMARY KEY
	username	VARCHAR(255) UNIQUE
	email	VARCHAR(255) UNIQUE
	password_hash	VARCHAR(255)
	created_at	DATETIME
Uploaded_Media	media_id	INT PRIMARY KEY
	user_id	INT FOREIGN KEY REFERENCES User(user_id)
	file_name	VARCHAR(255)
	media_type	VARCHAR(50)
	file_path	VARCHAR(512)
	uploaded_at	DATETIME
Detection_Results	result_id	INT PRIMARY KEY
	media_id	INT FOREIGN KEY REFERENCES Upload:_Media(edia(media_id))
	is_fake	BOOLEAN
	confidence_score	DECIMAL(5,2)
	heatmap_path	VARCHAR(512)
	uploaded_at	DATETIME
Frame_Metadata	frame_id	INT PRIMARY KEY
	media_id	INT FOREIGN KEY REFERENCES Uploaded:_Media(media_id)
	frame_number	INT
	frame_result	VARCHAR(50)

Figure 4.4: Database Design

The system uses a database (such as MongoDB) to store user inputs and detection results. The database structure is designed to efficiently manage and retrieve data.

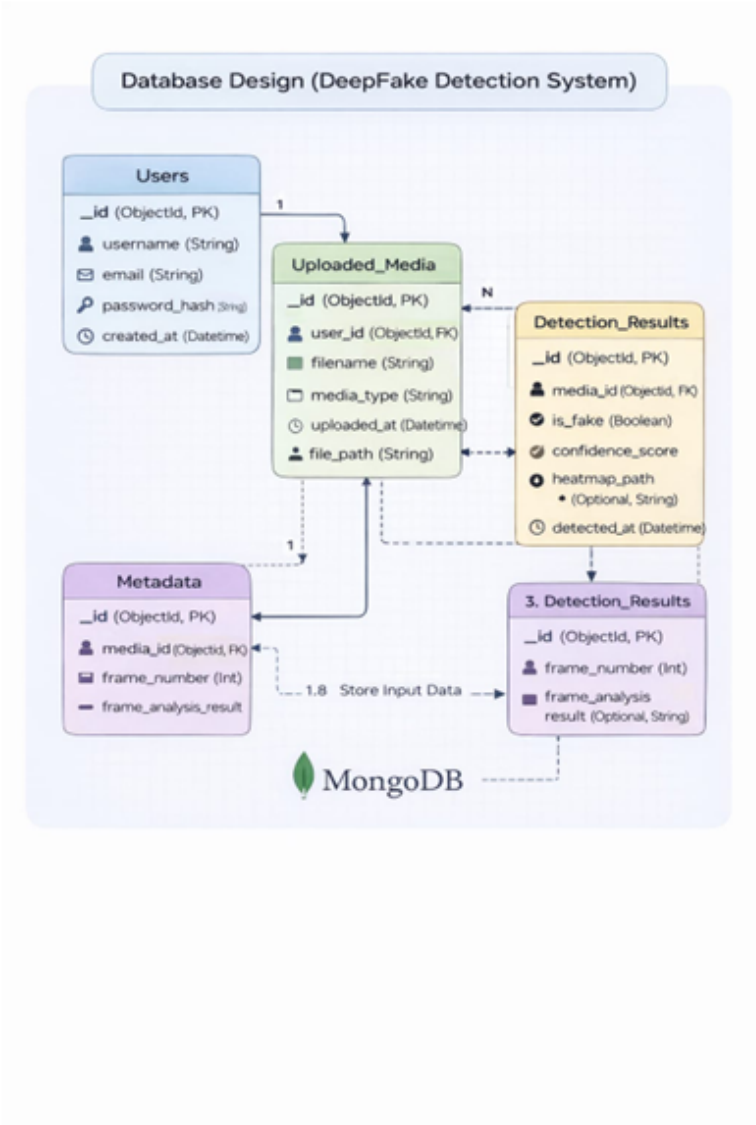

Figure 4.5: Database Design

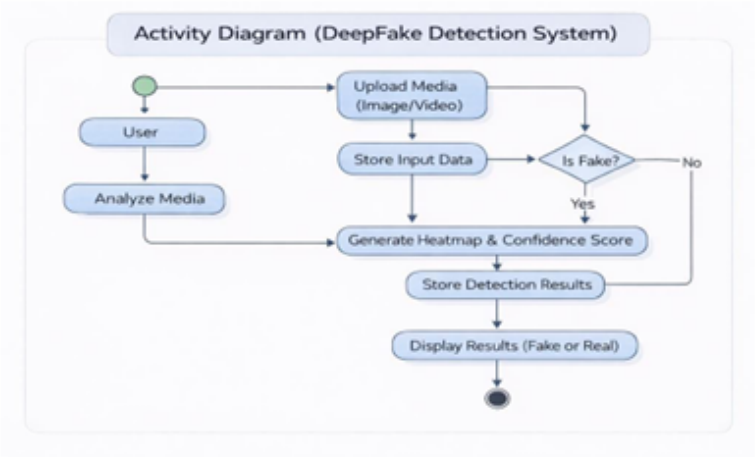

Figure 4.6: Activity Diagram

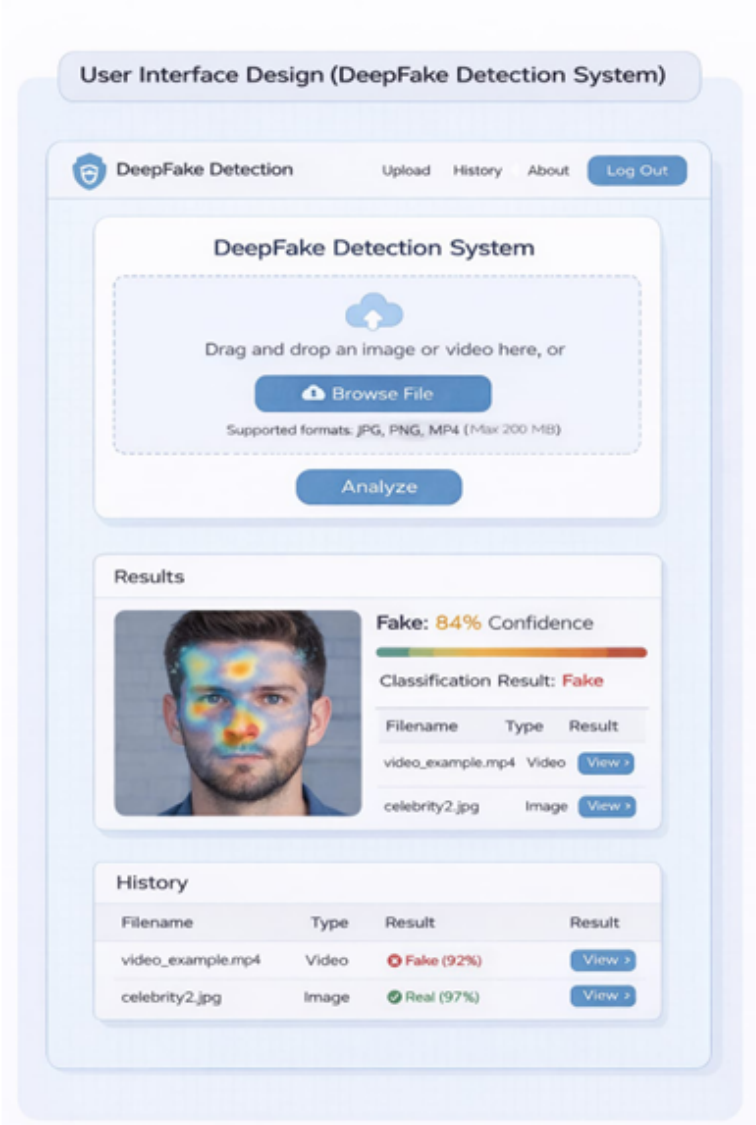

Figure 6.1: Frontend Interface

7. Generating predictions (Real/Fake)

8. Creating heatmaps using Grad-CAM

9. Sending results back to the frontend

The backend ensures smooth integration between machine learning models and the web interface.

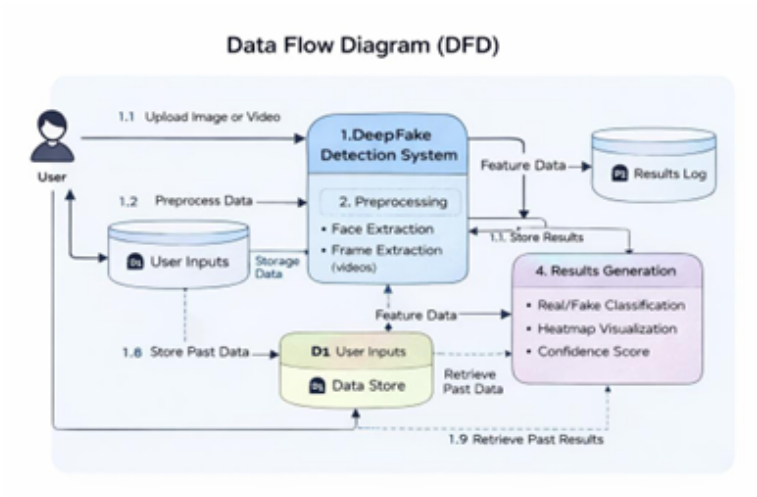

Figure 6.2: Backend Workflow

6.5 Web Scraping Implementation

To enhance the dataset and improve model performance, web scraping techniques can be used to collect additional images and videos from online sources.

- Tools such as BeautifulSoup and Selenium are used
- Data is collected from public platforms and datasets
- Scraped media is filtered and labeled
- Only relevant and high-quality data is stored

This process helps in expanding the dataset and improving the generalization of the model.

6.6 Data Processing and Insights

Data processing is a crucial step in ensuring accurate model performance. The collected data is cleaned, preprocessed, and prepared before training.

Steps involved: